

Long-term visual-to-tactile stimulation induces functional reorganization of thalamic pathways to achieve visual perception

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ABSTRACT

Sensory substitution devices (SSDs) have shown promising results in restoring basic visual function in blind subjects by translating visual information into tactile stimuli. However, the specific neuroplastic changes enabling this process remain poorly defined. Although cross-modal plasticity has been widely described through a range of intermediate cortical steps from somatosensory regions to the primary visual cortex, the brain's ability to search for a direct route makes the thalamo-occipital pathway the most viable reorganization option after long-term passive tactile stimulation. In this study, we use resting-state fMRI to investigate whether SSD training could redirect functional connections from the somatosensory thalamic nuclei to the primary occipital areas. After 6 months of training, blind children showed increased connectivity between the somatosensory thalamic nuclei and the occipital regions. This stronger connectivity was associated with improved performance in identifying tactile stimuli. These findings support the hypothesis that long-term passive tactile training leads to subcortical functional reorganization rather than cortical changes. Specifically, the results showed an increase in connectivity between somatosensory and multimodal integration thalamic nuclei and the visual cortical regions.

1. Introduction

Sensory substitution has enabled individuals who have lost a sense to perceive missing environmental information through an intact sense. In the case of blind individuals, vision loss is typically caused not by brain damage, but by issues with the pathways that transmit visual input to the brain. Sensory substitution devices (SSDs) are designed to transmit this missing input to the brain using another sensory modality, such as a tactile one. This technology exploits the brain's ability to reorganize itself and recruit brain areas responsible for the sense of sight to process visual information perceived by the sense of touch (Bach-y-Rita et al., 1969). However, the exact mechanisms underpinning this

tactile-to-visual perception remain poorly understood.

For both tactile and visual sensory information to reach the cerebral cortex, it is first processed through the thalamus. Tactile afferent input sends efferent projections to the ventral posterolateral nucleus (VPL), which subsequently targets the somatosensory cortex. For tactile information to reach the visual cortex, two main potential pathways exist: a cortico-cortical pathway passing through primary somatosensory areas, or an alternative more direct thalamo-occipital pathway. The reliance on either of these pathways may be shaped by the characteristics of the sensory stimulation received (Gilbert et al., 2009; Merzenich and Charms, 1996; Shmulevich and Kauffman, 2004; Szejka and Drossel, 2007; Villers-Sidani et al., 2010). Even so, how the brain reorganizes its

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thalamic pathways to the occipital lobe after long-term tactile stimulation training in blind individuals remains unknown.

Regarding the cortico-cortical pathway, several studies have demonstrated cross-modal connectivity from the primary somatosensory cortex (S1) to the primary visual cortex (V1) (Ioannides et al., 2013; Peltier et al., 2007; Wittenberg et al., 2004). Despite these studies supporting a strengthening in this pathway, the evidence suggests this route involves many intermediate cortical steps, including the secondary somatosensory cortex, the middle temporal visual area, the ventral intraparietal area and the parieto-occipital cortex (Négyessy et al., 2006), suggesting potential inefficiency.

In contrast, the thalamic-occipital pathway is hypothesized to be a more direct sensory route from the somatosensory thalamic nuclei to V1 cortex (Bieler et al., 2018; Henschke et al., 2015; Vasconcelos et al., 2011). While research in humans shows that tactile stimulation can activate the occipital cortex using connections originating from the lateral geniculate nucleus (LGN) (Müller et al., 2019; Reislev et al., 2017), animal studies have also shown that the visual cortex can receive input diverted from the somatosensory thalamic nuclei, like the VPL (Kaiserman-Abramof et al., 1980; Karlen et al., 2006). This finding suggests that the VPL may play a key role in subcortical functional reorganization to facilitate efficient cross-modal processing.

Although most of the literature to date has focused on tactile information reaching the occipital cortex via the cortico-cortical pathway, the potential for the brain to utilize a more direct route makes the thalamic-occipital pathway a compelling option for network-level changes following long-term passive tactile stimulation. In our current work, we set out to investigate if passive tactile stimulation encoding visual data can enhance functional connections from the somatosensory thalamus nuclei to the primary occipital areas. In blind children, we examined thalamic functional connectivity changes measured with fMRI after 6 months of training with SSD and assessed the association with visual identification performance. We hypothesized that somatosensory thalamic regions will show increased functional connectivity to cortical visual regions after SSD training, facilitating cross-modal neuroplastic changes.

2. Methods

2.1. Study design

We enrolled 24 children: 12 with blindness (5 girls and 7 boys, mean age = 9.9 years, SD = 1.1) and 12 sighted controls (5 girls and 7 boys, mean = 9.6 years, SD = 0.7), with ages ranging from 7 to 11 years. All children had similar academic performance and cultural backgrounds. The causes of blindness among children are described in Table 1. Participants were chosen among students from 12 randomly selected schools serving children with blindness in Madrid, Spain. Inclusion criteria for the participants were as follows: (a) ages between 7 and 11, (b) currently enrolled in school, and (c) consistent average academic performance, which was verified through grades and school

psychological reports. Exclusion criteria were: (a) having any sensorial deficit other than blindness, (b) having a current or past neuropsychiatric condition or (c) having a history of obstetric trauma resulting in cerebral hypoxia. The research protocol was approved by the Ethical Committee of Hospital Universitario Clínico San Carlos (Madrid) and was in full compliance with the Helsinki Declaration. Written information about the experiment was distributed to all participant schools. After principals and teachers approved the research protocol, a formal presentation was held at each school. Parents and teachers were given detailed information about the nature and purpose of the study, both verbally and in writing. Interested parents provided written consent after attending individual information sessions. Children were also given the same information verbally and were encouraged to ask questions during both group and individual sessions prior to providing their assent to participate in the study.

2.2. Tactile training

We use an SSD and an adapted training program, that allows blind individuals to recognize familiar objects from simple tactile (Bach-y-Rita et al., 1969). Tactile stimulation through the SSD consisted of a micro-camera (visual receptor) mounted on an eyeglasses frame transmitting images to a tactile stimulator (stimulation matrix) (Ortiz et al., 2011) (see Supplementary Figure 1). The device is 9 cm wide x 11 cm long x 1.8 cm high and composed of a controller and a vibrating pad. The core of the controller is a microprocessor that transforms images captured by the micro-camera into vibro-tactile impulses using ad hoc algorithms with features such as border processing, image stabilization, automatic contrast adjustment and separation of planes. It also controls the appropriate voltage and current for each pad's pixels. The $7 \times 7 \text{ cm}^2$ handcrafted vibrating pad is a two-dimensional array of 28×28 dielectric capacitors covered with a silicon layer. Inside the silicon layer and over each capacitor, there is a metallic ball to amplify the vibrating effect. Each stimulation point (metallic ball) represents a taxel (tactile pixel) with a diameter of 1.5 mm and is spaced 2.5 mm apart from one another. The minimum vertical movement of the actuators is 0.05 mm, and is refreshed every 35 milliseconds. The sweep signal of the entire image is performed every 75 milliseconds.

2.3. Behavioral training and training sessions

Blind children underwent twice-daily (morning and afternoon) training sessions, 5 days a week, to learn how to discriminate different stimuli using an SSD over a six month period. Each session lasted 30 min and consisted of systematic, orderly, and organized repetition of the visual stimuli transformed into tactile stimulation by the SSD. The content of each session changed during training. The first week's sessions consisted of displaying lines with varying spatial orientation (vertical, horizontal, and diagonal). Then, curves, simple silhouette forms, letters, words, and finally small sentences (up to 4–5 words each) were slowly incorporated. Additionally, two exercises using magnetic

Table 1

Basic demographics and cause of blindness of blind children participating in the study. F: Female, M: Male.

Subject	Age	Sex	Age at blindness	Light perception	Handedness	Cause of blindness
01	7	M	birth	no	right	Hereditary Retinal Dystrophy
02	8	M	6 years old	yes	right	Pigmentary Retinal Dystrophy
03	8	M	birth	yes	right	Pigmentary Retinal Dystrophy
04	9	F	birth	yes	right	Optic Disc Coloboma
05	9	M	birth	yes	right	Hereditary Retinal Dystrophy
06	9	M	birth	yes	right	Pigmentary Retinal Dystrophy
07	10	M	birth	yes	right	Pigmentary Retinal Dystrophy
08	10	F	birth	no	right	Anophthalmia
09	10	F	1 year old	yes	right	Optic Atrophy
10	10	F	birth	yes	right	Partial Optic Atrophy
11	11	F	birth	yes	right	Hereditary Retinal Dystrophy
12	12	M	3 years old	yes	right	Optic Atrophy

letters were included: (i) active touch recognition of the letters, and (ii) picking a specific letter among several options displayed on a white magnetic board. For each session, the average time needed to identify the visual stimuli correctly was recorded. A composite score reflecting the performance time for identifying different shapes and letters was computed to evaluate visual performance.

2.4. MRI acquisition

Blind children and their controls were scanned using a 3-T (ACHIEVA 3.0T TX; Philips) magnetic resonance imaging (MRI) scanner equipped with an eight-channel phased-array head coil. High-resolution 3D T1-weighted magnetization-prepared rapid acquisition gradient echo (T1W 3D TFE SENSE) images were acquired for anatomic reference (TR = 7.6 ms, TE = 3.5 ms, flip angle = 7μ , 1.0-mm isotropic voxels). Functional data were acquired using an echo planar imaging pulse sequence sensitive to BOLD contrast (TR = 3000 ms, TE = 30 ms, flip angle = 90μ , 3.0-mm isotropic voxels, sequence length = 6 min 12 s, 124 time points). Three functional sessions were acquired during each visit. In all cases, head motion was restricted using a pillow and several foam pads, and earplugs were used to attenuate scanner noise. During functional runs, participants were instructed to stay awake and remain as still as possible. Parents were allowed inside the scanner room to help their children stay still and feel comfortable.

2.5. MRI preprocessing

MRI data was preprocessed using FMRIB Software Library v5.0.7 (FSL) and MATLAB 2017a. The anatomical T1 preprocessing pipeline included: reorientation to right-posterior-inferior (RPI); alignment to anterior and posterior commissures; skull stripping; gray matter, white matter and cerebrospinal fluid segmentation; and computation of non-linear transformation between individual skull-stripped T1 and 2 mm resolution MNI152 template images. The functional MRI preprocessing pipeline included: slice time correction; reorientation to RPI; realigning functional volumes within runs with rigid body transformations (6 parameters linear transformation); computation of the transformation between individual skull-stripped T1 and mean functional images; intensity normalization; removal of confounding factors from the data using linear regression -including 24 friston motion-related covariates (6 rigid motion parameters, their derivatives and the square of these 12 parameters to account for possible quadratic effects), linear and quadratic terms, and five components each from the lateral ventricles and white matter. Of note, global signal regression was not applied due to the spurious correlations this can introduce. Transformation of resting-state data to MNI space, concatenating the transformation from functional to structural and from structural to MNI, spatial smoothing with an isotropic Gaussian kernel of 6-mm FWHM, and band-pass filtering (0.01–0.08 Hz) to reduce low-frequency drift and high-frequency noise were also performed. Head motion was quantified using realignment parameters obtained during image preprocessing, which included 3 translation and 3 rotation estimates. Scrubbing of time points with excess head motion eliminated all time points with a frame displacement > 0.5 mm. The 3 resting-state runs were concatenated and after removing time points exceeding the threshold, we obtained 326 time point series. The distributions of the correlations across all-time series were inspected for possible noise contamination; no outliers were observed from the whole-brain connectivity distributions across all participants.

2.6. Voxel-level thalamic functional connectivity

To evaluate thalamic functional reorganization, the individual subject's thalamic connectivity matrix was computed. Pearson correlation was used to derive high-resolution functional connectivity between each thalamic voxel (1047 voxels) and all other voxels in the brain (49,875

voxels), yielding a $1047 \times 49,875$ connectivity matrix.

2.7. Thalamic connectivity reorganization (Weighted degree maps)

First, to evaluate the global functional connectivity differences between the thalamus and the cortex, we computed weighted degree maps. This process yielded a whole-brain map where each cortical voxel represents the total functional connectivity strength between that cortical voxel and all voxels within the thalamus. A voxel-level mass univariate two-sample unpaired *t*-test was used to compare the thalamic connectivity between blind children and sighted controls, adjusting for age and mean frame displacement, and a single-group paired *t*-test was used to study post-training effects within the blind cohort. Results were corrected for multiple comparisons using a cluster-wise correction based on Monte Carlo simulation using the 3dClustSim tool (AFNI software; <https://afni.nimh.nih.gov/>) with 10,000 iterations, $\alpha = 0.05$ and *p* value < 0.05. This procedure estimated the spatial autocorrelation of the residual data and generates 10,000 random maps with the same autocorrelation to determine the minimum cluster size (contiguous group of uncorrected significant voxels) required to achieve Family-Wise Error (FWE) correction at *p* < 0.05.

2.8. Thalamic nuclei to occipital connectivity reorganization (Link-Level analysis)

In a similar fashion, a mass univariate General Linear Model (GLM) was used to perform the statistical tests of interest, but in this case, for each individual functional connection (link) between a thalamic voxel and a cortical voxel. To ensure the robustness of these specific thalamo-cortical connections, we applied a stringent link-level whole-brain correction for multiple comparisons, adapting the Monte Carlo simulation method to link-level analysis (Diez et al., 2020; Guzmán-Vélez et al., 2022). This approach requires that both the seed (thalamic) and the target (cortical) clusters of the connection survive stringent cluster-wise correction, as recommended for brain-wide association statistical testing (Gong et al., 2018). Specifically, we first used a cluster-wise Monte Carlo simulation within the cortex for each thalamic voxel's connectivity map (10,000 iterations, $\alpha = 0.05$ and *p* value < 0.05) to determine the minimum required cortical cluster size. We then performed a separate cluster-wise Monte Carlo simulation within the thalamus to determine the minimum required size for a thalamic seed cluster. While the simulation indicated a minimum threshold of approximately 40 voxels within the thalamus for FWE correction, we conservatively applied a larger threshold of 200 contiguous voxels to ensure that effects implicated larger, functionally coherent thalamic regions and increased anatomical interpretability.

2.9. Association with visual performance

Additionally, a multiple regression model was used to remove the effect of age and mean frame displacement and study the association between visual identification performance after six months and the change in thalamic-to-occipital functional connectivity. The same link-level network-based correction for multiple comparisons was applied to this association analysis to ensure statistical validity.

2.10. Thalamic nuclei connectivity to visual regions

To further distinguish those thalamic alterations involved with visual information processing, only the surviving functional links ending in occipital regions were selected. The *Freesurfer* probabilistic atlas of thalamic nuclei was used to identify the involved thalamic nuclei (Iglesias et al., 2018). The number of surviving thalamic connections ending in V1, V2, V3, V4, and V5 after training was quantified. Additionally, thalamic parcellation based on 7 resting state Yeo networks (Raut et al., 2020; Yeo et al., 2011) (<https://github.com/ryraut/thalam>

3. Results

3.1. Thalamic functional connectivity reorganization at baseline scan in blind children

2). Additionally, blind children did not show increased thalamic connectivity compared to the sighted controls.

3.2. Thalamic functional connectivity reorganization in blind children after SSD training

We evaluated thalamic functional connectivity reorganization in blind children after training with an SSD by computing a paired *t*-test comparing baseline and 6-months scans. Our findings indicated that there were relative increases in thalamic functional connectivity with cortical visual regions, the anterior cingulate cortex, anterior insula, basal ganglia and brainstem nuclei after the training (Fig. 1B; **Supplementary Fig 2**). No significant decreases in thalamic connectivity was observed in blind children after training. Additionally, we employed an unpaired two-sample *t*-test to further analyze the differences in thalamic functional connectivity between blind children who underwent SSD training and sighted controls. This analysis revealed that in blind children, the anterior-medial and lateral aspects of the occipital cortex exhibited decrease connectivity. In contrast, the posterior-medial aspect of the visual regions showed increased connectivity to the thalamic regions when compared to the sighted controls (Fig. 1C; **Supplementary Fig 2**).

3.3. Specific thalamic nuclei to occipital functional connectivity reorganization

To identify the discrete thalamic nuclei displaying altered functional connectivity to the occipital regions, links connecting both the thalamus and the occipital region were evaluated. Prior to SSD-training, blind children showed reduced connectivity from the lateral geniculate (LGN) and medial geniculate (MGN) nuclei, ventral lateral nucleus (VL), pulvinar lateral nucleus (PUL) and mediodorsal medial magnocellular

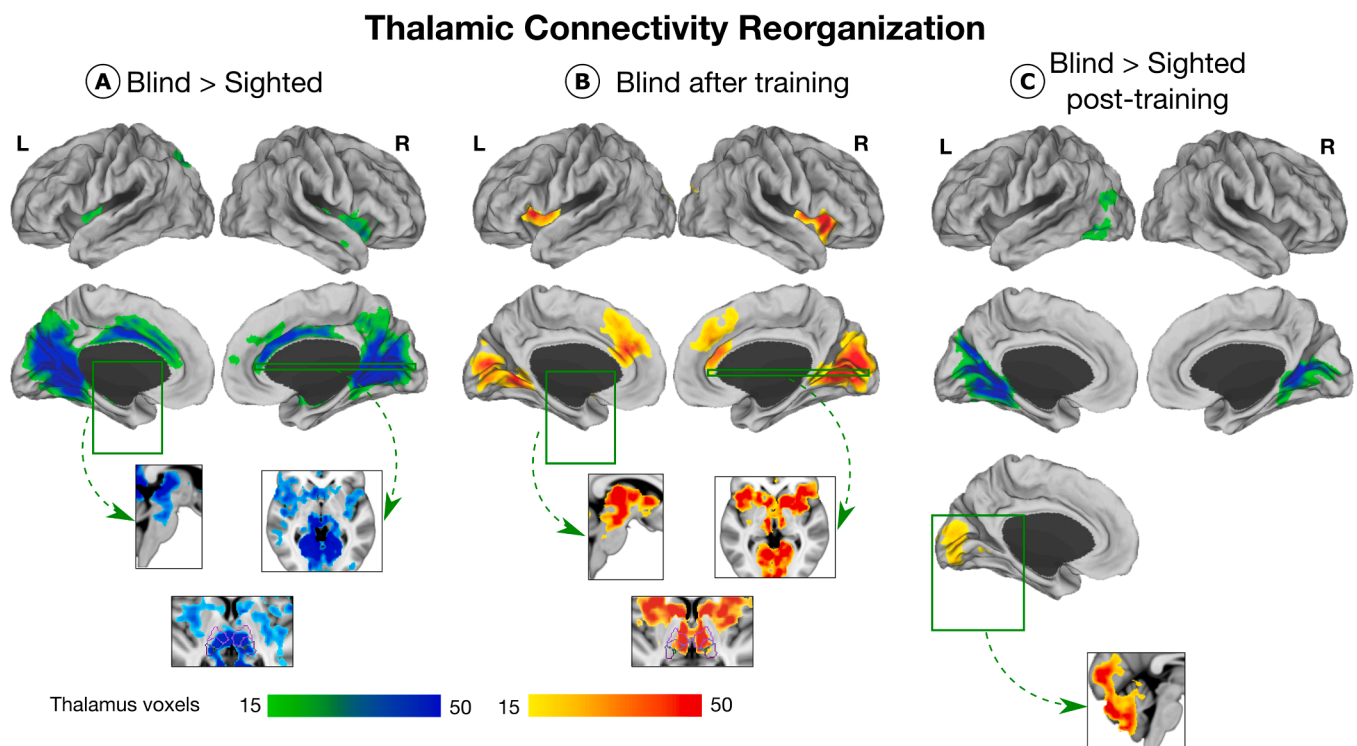


Fig. 1. Thalamic connectivity in blind children and its reorganization after SSD training. The visualization highlights voxel-wise differences in thalamic functional connectivity between pairs of groups. **(A)** Blind children showed reduced thalamic connectivity to the cingulate cortex, precuneus, insula, anterior basal ganglia, brain stem, and visual regions compared to sighted controls. **(B)** After training, blind children show increased thalamic connectivity with visual regions, anterior cingulate cortex, anterior insula, basal ganglia, and brainstem. **(C)** Blind post-training children showed increased thalamic connectivity to the posterior-medial aspect of the visual regions and reduced connectivity to the anterior-medial and lateral aspects compared to sighted controls. Axial view of the thalamic regions displays the boundaries of the Yeo 7 functional networks (Yeo et al., 2011).

nucleus (MDm) to occipital regions when compared to sighted controls (Fig. 2A; **Supplementary Fig 3**). After completing the training, blind children showed increased connectivity from the ventral posterolateral (VPL), ventral lateral posterior (VLP) and pulvinar anterior nuclei (PuA) to the occipital regions. Most of these significant connections were observed to V1, with the connectivity progressively decreasing toward V5 (Fig. 2A; **Supplementary Fig 3**). Despite the training, blind children still exhibited reduced functional connectivity from the LGN, MGN and MDm, but had increased connectivity from the VL and medial pulvinar (PuM) compared to sighted controls. Sighted controls of implicated thalamic nuclei is shown in **Supplementary Fig 4**.

3.4. Thalamic connectivity reconfiguration in the blind children after training

To characterize the cortical functional reorganization of thalamic nuclei that exhibit increased connectivity after training, seed-based analyses of these nuclei were performed. At baseline, these thalamic regions showed strong connectivity to somatosensory and multimodal integration cortices (ventral attention network). After training, the functional connectivity of these nuclei connectivity further extended to include visual regions (Fig. 2B).

3.5. Thalamic connectivity reorganization associated with individual differences in performance after training

A regression analysis was conducted to find associations between individual differences in visual performance, measured by SSD after training, and thalamic functional connectivity. Children who exhibited better visual performance showed increased functional connectivity between the VPL and the visual areas (Fig. 3; **Supplementary Fig 5**).

3.6. Increased thalamic-occipital functional connectivity after training

To better understand whether the functional rerouting of tactile information to visual areas was driven by a cortico-cortical pathway (which involves the VPL nuclei in the thalamus – cortical somatosensory region – intraparietal area – visual regions) or a thalamic-occipital pathway (which includes the VPL nuclei in the thalamus – LGN in the thalamus – visual regions), we computed connectivity differences between aforementioned seeds (Fig. 4). After training, blind children showed a significant increase in functional connectivity from both the VLP and VLP nuclei to the V1 region (p -value=0.0002 in both cases). However, the connectivity from the LGN to V1 did not significantly increase after training. This may be due to the initial high connectivity value that already existed between these two regions. Before training, LGN to V1 connectivity was significantly lower in blind children compared to sighted controls, but this difference was not significant when comparing the connectivity of blind children after training. When examining the cortico-cortical pathway, no significant changes were observed in the connectivity from the VPL to S1 from S1 to intraparietal regions. The connectivity from the intraparietal region to V1 showed increased connectivity in blind children before training compared to sighted controls; however, this connectivity significantly decreased after training (p -value=0.03).

4. Discussion

In this study, we present a novel finding showing that continuous training with the Sensory Substitution Device (SSD) induces functional subcortical reorganization that supports an efficient transition from sensation to tactile perception in blind children. The SSD transforms visual input into tactile stimuli that are transmitted through discrete stimuli, encoded discontinuously in space and time, and are subsequently converted into passive tactile perception (Barrea et al., 2018; Schwarz, 2016). Our results suggest that simple tickling sensations,

initially generated by isolated stimuli in the palm of the hand, evolve into meaningful visual-like perceptions as a network-level functional reorganization occurs once a six-month training is completed. Specifically, we observed that blind children exhibit enhanced functional coupling from the somatosensory thalamic nuclei, particularly the VPL, to the occipital cortex after long-term passive tactile stimulation, when compared to sighted controls. Furthermore, individual differences in performance are significantly correlated in a 'dose-dependent type fashion' with the post-training VPL-occipital cortex functional connectivity strength, underscoring the functional relevance of this reorganized pathway.

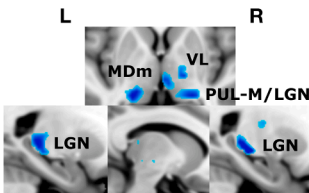
The plasticity of the visual cortex is influenced by external stimulation, which depends on the functional maturation of local inhibitory connections (Henschke et al., 2015). This means that the strengthening of structural and functional brain connections is experience-dependent (Ismail et al., 2017). In blind individuals, the absence of visual input leads to widespread cortical (Ortiz-Teran et al., 2017) and thalamo-cortical (Marins et al., 2023) pathways reorganization. However, our research reveals that long-term tactile-to-visual stimulation using SSD is capable of reorganizing functional connectivity, creating a direct functional pathway to the V1 via the VPL. This direct connection has previously only been observed in bilaterally enucleated animals (Karlen et al., 2006). We suggest that the reason lies in the type of stimulation employed. Instead of the gradually displayed stimulation (e.g., several second timescale) typically used by blind individuals when reading Braille, our approach allows them to perceive the stimuli in milliseconds, in a similar fashion to how sighted individuals process visual information as a whole.

The functional reorganization of the VPL-occipital pathway may be explained by how a specific repetitive passive tactile stimulation can gain predominance over other inputs while determining the value of the output (Kauffman et al., 2004; Szejka and Drossel, 2007) needed by the visual cortex. Within the characteristics of these sensory inputs, such as the type, magnitude, duration and interval between stimuli, resides the possibility of redirecting the neural response (Gilbert et al., 2009; Kilgard et al., 2007; Merzenich and Charms, 1996; Villers-Sidani et al., 2010), as well as the timing in which the stimulation is carried out, especially in sensitive periods, where the mechanisms of cross-modal neuroplasticity are highly activated (Büchel et al., 1998; Burton et al., 2002; Knudsen, 2004; Voss, 2013). Furthermore, it has been suggested that the first 16 years of life represent a crucial period for a functional shift of V1 from processing visual stimuli to processing tactile ones (Sadato et al., 2002). Consequently, to effectively strengthen the thalamic-occipital pathway, the duration of the stimulation must extend beyond these critical periods, as Ramón y Cajal (1904) highlighted this point when he stated that "the acquisition of new skills requires many years of mental and physical practice for the formation of new pathways through ramification and progressive growth of the dendritic arborization and the nervous terminals."

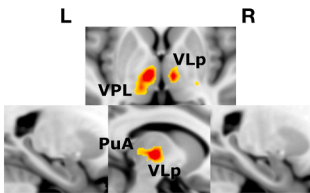
The study of thalamic functional reorganization in the blind brain is challenging. Previous high-precision studies, particularly in pharmacological fMRI investigating compounds such as LSD (Pizzi et al., 2023) and Modafinil (Pizzi et al., 2025), have successfully used anatomical parcellation of thalamic nuclei from structural images. These nuclei are then used as seeds to perform voxel-wise connectivity analysis. This approach yields excellent results because the participants are healthy individuals with presumed stable functional anatomy, allowing the study of acute, chemically induced functional changes. However, our initial thalamic connectivity analysis comparing blind children with sighted controls (Fig. 1) suggested profound neuroplastic changes in the blind thalamus. These changes likely occur early in life due to the brain's ability to reorganize and cope with the absence of visual stimuli. This potential alteration of typical thalamic nuclei boundaries led us to develop an unbiased, data-driven approach. We therefore focused on thalamic voxel-wise connectivity instead of using a priori seeds derived from standard (sighted control) anatomical atlases, which allow us to

A Thalamus to Occipital Connectivity

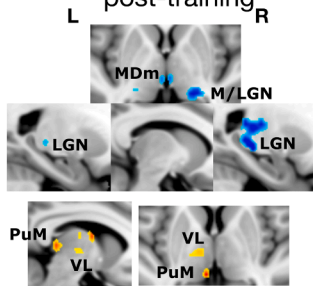
Blind > Sighted



Blind after training

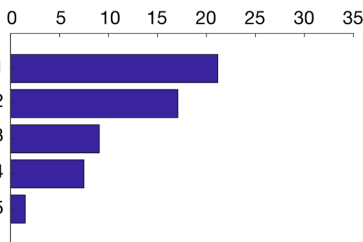


Blind > Sighted post-training

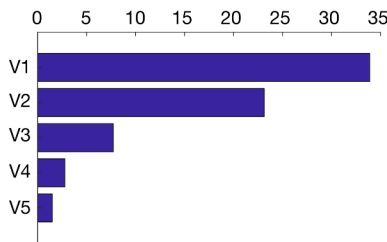


Occipital voxels 200 1000 200 1000

Left hemisphere



Right hemisphere

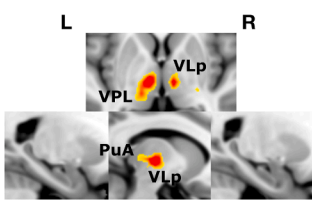


Number of voxels in Thalamus connecting to visual regions after training.

Visual regions

B Thalamus connectivity reorganization after training

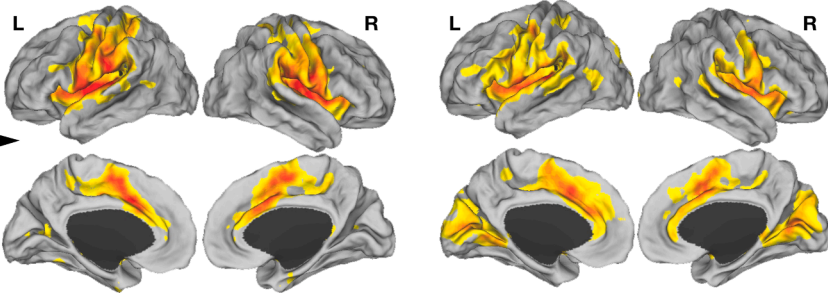
Blind after training



200 1000

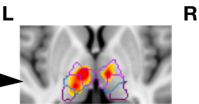
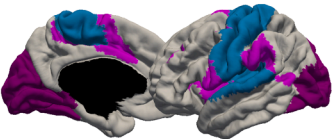
Thalamus increased connectivity to Occipital after training

Connectivity reconfiguration from statistical significant thalamic regions
Blind Baseline Blind after training

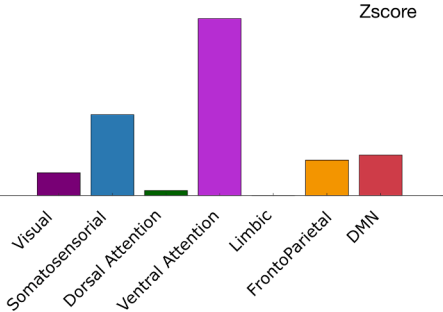


0.5 2.5
Zscore

Yeo functional parcellation in Thalamus



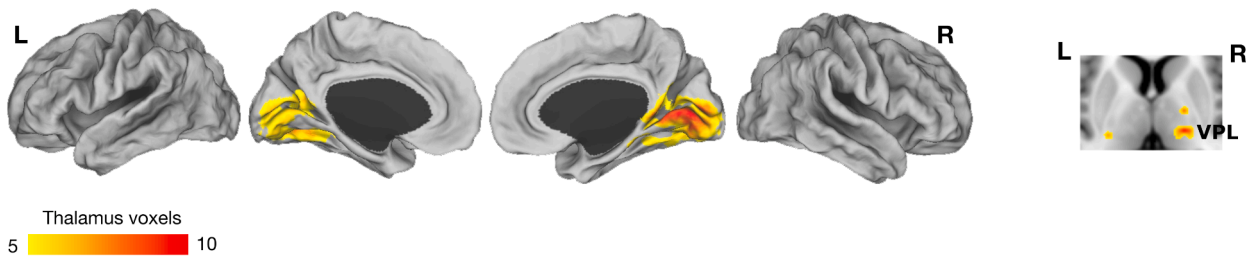
Thalamus voxels in Yeo ROIs of thalamus



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Fig. 2. Thalamic connectivity reconfiguration. (A) Thalamic regions with significantly increased/decreased functional connectivity to the occipital cortex are displayed. Top row: Differences in connectivity displayed on axial and sagittal slices. Left: Blind > Sighted before training, showing areas of decreased connectivity in blind children (cool colors). Middle: Connectivity changes in Blind after training, showing areas of increased connectivity (warm colors). Right: Blind after training > Sighted, showing areas of greater connectivity in blind children after training (warm colors) and decreased (cool colors) compared to sighted controls. Color bar indicates the number of voxels. Bottom row: Bar plots illustrating the number of significant voxels within different visual regions (V1-V5) showing altered connectivity. The left bar plot corresponds to the left hemisphere, and the right bar plot to the right hemisphere. (B) Thalamic connectivity reorganization after training is displayed. Top row: Cortical functional connectivity maps of the thalamic cluster that showed increased connectivity after training (as shown in panel A, 'Blind after training' connectivity map). The connectivity patterns are displayed at Blind Baseline and Blind after training. Color bars indicate Z-transformed mean functional connectivity. Bottom left: Visualization of the boundaries of the Yeo 7 functional networks (Yeo et al., 2011) in the thalamus overlaid on the significant thalamic regions and the implicated cortical regions projected to the cortex. Different colors represent distinct functional networks. Bottom right: Histogram depicting the distribution of significantly reconfigured thalamic voxels across the different Yeo functional networks (Visual, Somatosensory, Dorsal Attention, Ventral Attention, Limbic, Frontoparietal, DMN). The y-axis represents the count of thalamic voxels for each network. Key thalamic nuclei are indicated: lateral geniculate nucleus (LGN), medial geniculate nucleus (MGN), pulvinar (PUL), ventral lateral nucleus (VL), ventral posterolateral (VPL), ventral lateral posterior (VLP), mediodorsal medial magnocellular nucleus (MDm), pulvinar anterior (PuA), and medial pulvinar (PuM).

Thalamic FC correlated with performance in 6 months



Higher connectivity of right VPL with visual regions after training is associated with better performance

Fig. 3. Thalamic connectivity reconfiguration associated with visual performance after training. Regions showing reconfigured thalamic-cortical connections correlated with visual performance after training are shown. Children with better visual performance showed an increased functional connectivity between the VPL and the visual areas.

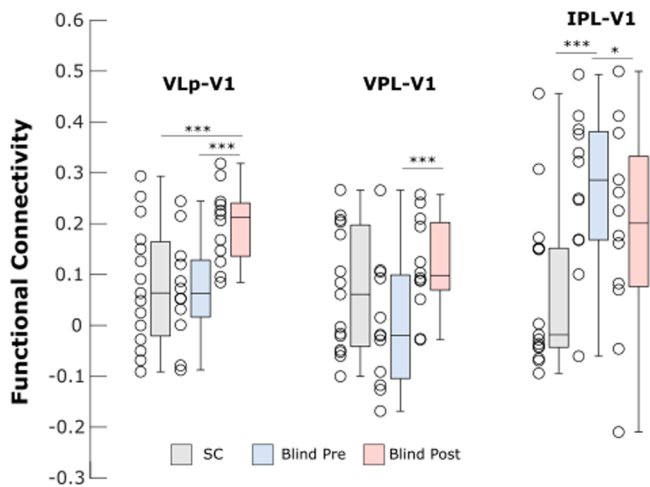


Fig. 4. Increased thalamic-occipital pathway connectivity after training. Boxplots of significant connectivity changes in blind children after training between key seeds in thalamic-occipital and cortico-cortical pathways. After training, blind children showed increased connectivity from VLP and VPL to V1 regions (thalamic-occipital pathway), and decreased connectivity between the intraparietal region and V1 (cortico-cortical pathway).

capture the functional organization reorganization both inside and outside of the thalamus. While an anatomical seed of the nuclei would be the preferred choice for a brain that does not expect substantial neuroplastic changes, in this case we believe this approach can better capture the unique alterations of the blind brain.

Thalamic-cortical neuroplasticity occurs not only in cortical areas but also at the somatosensory thalamic level, where visual and tactile responses converge (Allen et al., 2017). This neuroplasticity is observed in the same areas as in sighted human controls (Raut et al., 2020). Our results revealed a fast, robust functional connectivity in the VPL towards the V1, which reduces intermediate cortical steps, thereby enhancing speed and efficiency after the training. Thalamus nuclei reorganization produces advantageous changes in response to an environment that will help the system to fulfill its goals (Gershenson, 2012). Other researchers have also found similar results through the LGN to the occipital cortex (Müller et al., 2019) using non-visual stimuli, based on the structural and functional connectivity of the LGN and optic radiations relaying information to the V1 (Cecchetti et al., 2016). The LGN-V1 pathway has been suggested to occur as a result of the connections between auditory and somatosensory thalamic nuclei (Karlen et al., 2006; Rehkämper et al., 1994), as the thalamus plays a role in selecting, organizing, and directing stimuli toward the cerebral cortex. Given that in blind individuals this pathway is stunted (Noppeney et al., 2005; Shimony et al., 2006; Shu et al., 2009), cross-modal plasticity within the VPL can bypass the traditional retinofugal pathway (from the LGN to the V1), even though the VPL is directly connected to the somatosensory cortex (Padberg et al., 2009). Additionally, the hand receptive fields are associated with the somatosensory parietal area (Krubitzer et al., 2004).

There are a few limitations in this paper, such as the small sample size and the etiological heterogeneity of the cause of peripheral blindness. However, we have attempted to address these issues by conducting a lengthy follow-up and constant work with the children in the study for a six-month period, and cutting-edge analysis methods. This study serves as a critical exploratory investigation—a necessary first step before pursuing larger, stratified cohorts—to confirm the feasibility and

efficacy of the SSD training in driving functional reorganization and promoting perception through touch across different peripheral etiologies. A key limitation in the study design is the absence of a blind, non-trained control group, which prevents the definitive disentanglement of training-specific effects from general developmental or blindness-specific effects over the six-month period. Future studies with larger and stratified samples and non-trained blind group will be necessary to fully map the common and distinct patterns of neuroplasticity resulting from different etiologies. Additionally, while the focus of this present study pertained to *a priori* thalamo-occipital hypotheses, post vs. baseline training increased connectivity was also observed between the thalamus and cingulo-insular (salience network) brain areas (Seeley et al., 2007; Sepulcre et al., 2012). Although positive correlations between VPL-occipital cortex connectivity and post-training performance underscores the biological importance of this specific pathway, additional research is needed to further contextualize the role of cortical multimodal integration brain areas in cross-modal experience-dependent neuroplasticity. Furthermore, it remains unclear whether the identified functional connectivity alterations following SSD training also reflect structural connectivity changes. These questions will need to be investigated in a future cohort with available diffusion-weighted imaging data.

We conclude that the biological significance of the somatosensory thalamus-cortical pathway, achieved through long-term passive tactile stimulation, lies in the direct communication of somatosensory information from the VPL to the occipital cortex without the need for intermediate steps. The ability of the thalamus to reorganize its rapid pathways depends on the duration and type of tactile stimulation it receives.

Data availability

All data and codes related to imaging analysis are available for the research community upon request for the purpose of scientific investigation, teaching, or the planning of clinical research studies

CRediT authorship contribution statement

Elena Ortiz-Teran: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Laura Ortiz-Teran:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation. **David L Perez:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation. **Tomas Ortiz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ibai Diez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2025.121652.

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